基本放大电路

峰值/幅值 magnitude amplitude

峰峰值 peak to peak amplitude

电信号 { 小电压 / 小电流

声音 I_i → 放大电路 含电源 → I_o Load 负载 → 声音 (扩大声音)

U_i 小对峰 2mV → U_o 大对峰 100mV

放大电路基本特征：

① 能够实现功率的放大

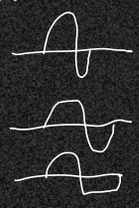
$\frac{U_o}{I_i}$ 互阻放大倍数 Ω

$\frac{I_o}{I_i}$ 电流放大倍数

$\frac{I_o}{U_i}$ 互导放大倍数 $\frac{A}{V}=\frac{1}{\Omega}=S$ 西门子

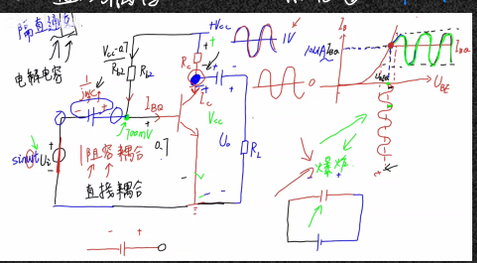
$\frac{U_o}{U_i}$ 电压放大倍数

失真：

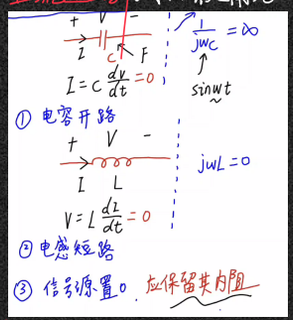


$I_C R_C + U_o = V_{CC}$

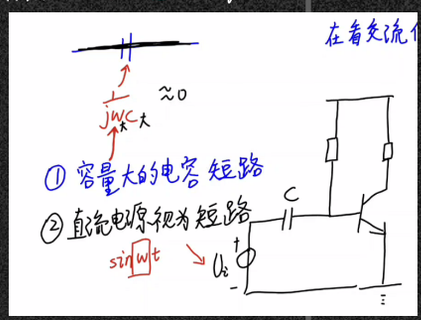
直接耦合 阻容耦合 (两种实用放大电路)



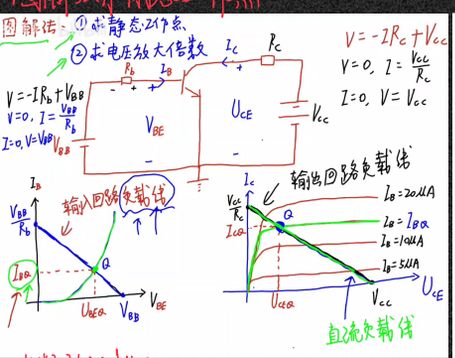
直流通路：用于研究静态工作点 (quie scent)



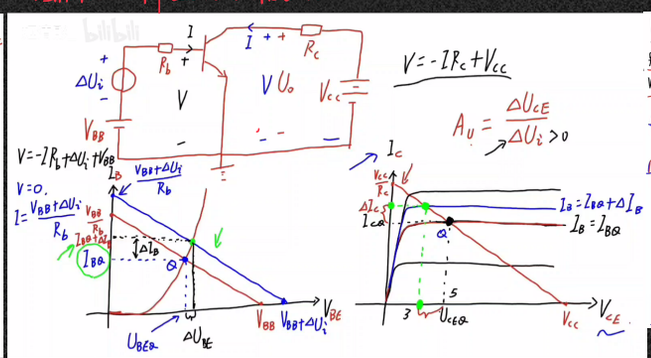
交流通路：用于研究动态参数

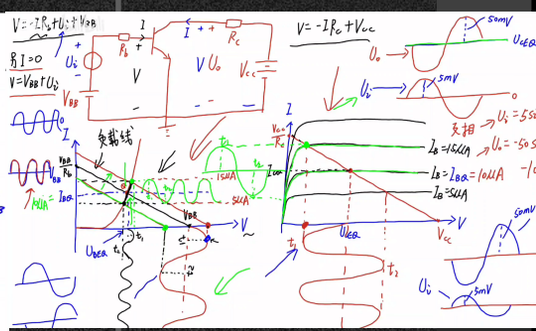


图解法求静态工作点

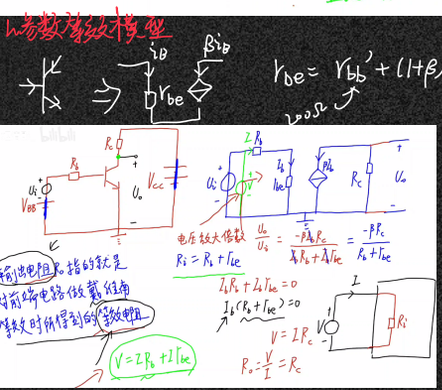


图解法分析电压放大倍数



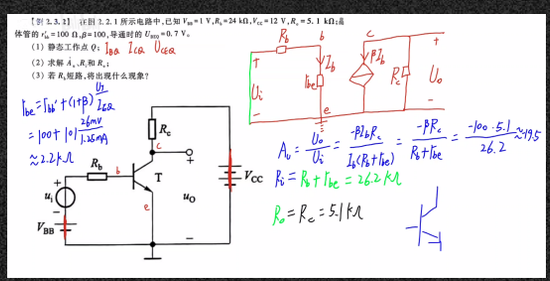


h参数等效模型



$r_{be} = r_{bb}' + (1+\beta)\frac{U_T}{I_{EQ}}$ ← 26mA

$I_{EQ} = (1+\beta) I_{BQ} \approx I_{CQ}$



【例 2.3.2】 在例 2.2.1 所示电路中，已知 $V_{BB}=1\ V$，$R_b=24\ k\Omega$，$V_{CC}=12\ V$，$R_c=5.1\ k\Omega$；晶体管的 $r_{bb}'=100\ \Omega$，$\beta=100$，导通时的 $U_{BEQ}=0.7\ V$。

(1) 静态工作点 Q：I_{BQ} I_{CQ} U_{CEQ}

(2) 求解 A_u、R_i 和 R_o；

(3) 若 R_b 短路，将出现什么现象？

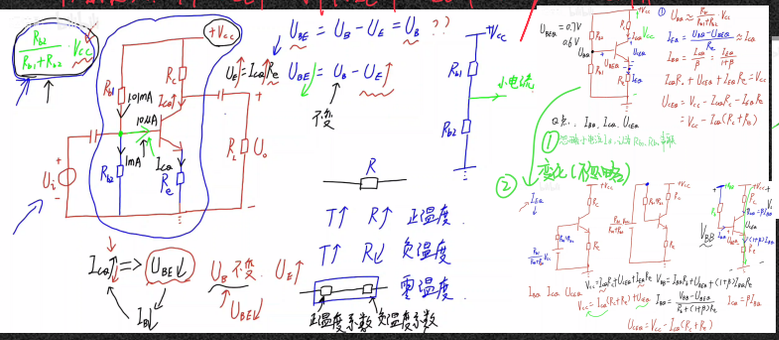
$I_{be} = r_{bb}' + (1+\beta)\frac{26mV}{I_{EQ}} = 100 + 101\frac{26mV}{1.15mA} \approx 2.2k\Omega$

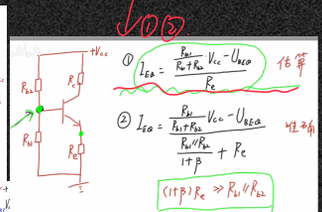
$A_u = \frac{U_o}{U_i} = \frac{-\beta I_b R_c}{I_b(R_b + r_{be})} = \frac{-\beta R_c}{R_b + r_{be}} = \frac{-100 \cdot 5.1}{26.2} \approx -195$

$R_i = R_b + r_{be} = 26.2k\Omega$

$R_o = R_c = 5.1k\Omega$

静态工作点的稳定 $T\uparrow \rightarrow I_C\uparrow$ 抑制 $I_C\uparrow$ $I_C = \beta I_B$ / Q点的计算





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